

Monochrome Camera Based Fluorescence Spectroscopy Device for Cancer Diagnosis

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Abstract: We have developed a compact, portable, and automated polarized fluorescence spectroscopy device using a monochrome camera and a Raspberry Pi module. The system enables accurate spectra in comparison with a commercial spectrometer, demonstrating potential for oral and cervical cancer diagnostics.

1. Introduction

Fluorescence spectroscopy is a rapid, noninvasive, and quantitative technique for assessing biochemical and morphological changes associated with cancer development [1,2]. Our group previously developed smartphone-based fluorescence spectroscopy devices for early detection of oral and cervical cancer [3,4]. While these devices demonstrated portability, affordability, and the advantages of smartphone technology, smartphone image sensors have inherent limitations. The presence of color filter arrays (CFAs) leads to (i) spectral undersampling and (ii) reduced sensitivity due to light attenuation. Furthermore, poor standardization across models results in variations in sensor characteristics among different smartphones. To address these challenges, we propose a monochrome camera-based fluorescence spectroscopy device. Monochrome sensors eliminate color filter artifacts, provide uniform spectral response, and offer higher sensitivity and a better signal-to-noise ratio. The compact and low-cost platform supports automated operation and real-time analysis, offering potential applications in cancer diagnostics.

2. Materials and Method

We have designed and developed a monochrome camera-based fluorescence spectroscopy device utilizing a 405 nm laser source for sample illumination. The optical setup of the device is shown in Fig. 1(a). A linear polarizer was placed in the illumination path to polarize the incident light, and another polarizer (analyzer) in the detection path captured fluorescence signals at parallel (Co) and perpendicular (Cross) polarization states. A rotation motor with a gear module and position counter enabled precise and repeatable alignment of the analyzer. A 450 nm long-pass filter blocked excitation light, ensuring accurate fluorescence signal detection. The filtered fluorescence light was subsequently dispersed using a transmission grating, and the resulting spectrum was captured by a high-sensitivity monochrome sensor. The complete system is powered by a Raspberry Pi 4 board running RPi OS (Bullseye), which manages all device operations. The board's GPIO pins regulate the illumination intensity via a digital potentiometer. The entire setup is fully automated to capture spectral images under linear polarization configurations. Figure 1(b) presents the device automation flowchart, which illustrates the automated control of user-defined illumination intensities, power on/off switching, source selection, and image acquisition. The captured spectral data are processed using Python libraries for fluorescence signal analysis. This automation framework minimizes operator subjectivity, enhances measurement repeatability, and simplifies overall device operation. The device was modeled using SolidWorks software and fabricated through 3D printing (fused deposition modeling) using a biocompatible and non-fluorescent medical-grade polylactic acid (PLA).

3. Results and Discussion

The device captures fluorescence signals as 8-bit raw images. Each pixel in the spectral image corresponds to a specific wavelength of the emitted light. To extract fluorescence intensities corresponding to these wavelengths, pixel-to-wavelength calibration was performed using a mercury lamp with multiple sharp emission peaks. Figure 2(a) shows the emission spectrum of the mercury lamp captured using the developed device, alongside the reference spectrum obtained from a commercial Ocean Optics spectrometer. The comparison demonstrates that the

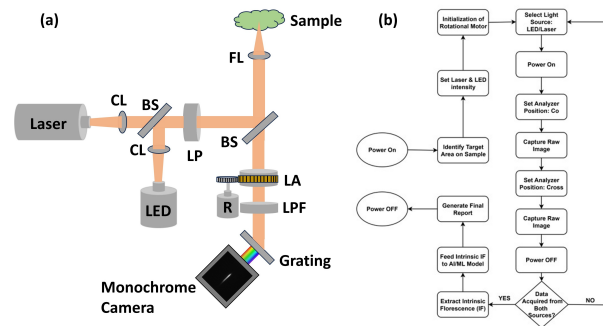


Fig. 1. (a) Schematic of the optical setup of the monochrome camera-based developed device (CL: collimating lens; BS: beam splitter; LP: linear polarizer; LA: linear analyzer; R: rotation motor; LPF: long-pass filter; FL: focusing lens). (b) Flowchart illustrating the device automation process.

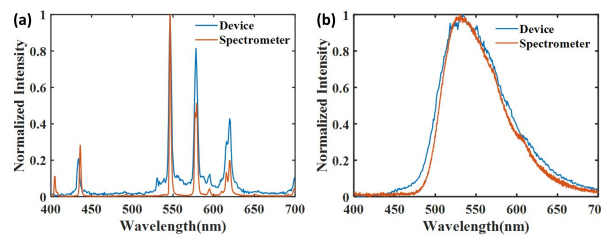


Fig. 2. Emission spectra captured using the developed device and a commercial spectrometer: (a) Mercury lamp, (b) FAD liquid phantom.

device efficiently captures all major emission peaks of the mercury lamp within the 400–700 nm range. The calibrated system was further validated in the laboratory to assess its performance in detecting fluorescence signals. A flavin adenine dinucleotide (FAD) liquid phantom was used as a standard fluorescent sample, and its emission spectrum was recorded using both the developed device and the commercial spectrometer under identical conditions. The resultant spectra shown in Figure 2(b) depict the efficiency of the present device to capture fluorescence signal in comparison with the spectrometer. Both spectra have the same peak position of FAD (525 nm) and spectral broadening, especially near the peak position. These preliminary results confirm the capability of the monochrome camera sensor to capture fluorescence spectroscopy signals with high accuracy and precision.

4. Conclusion

The use of a monochrome camera sensor enables the capture of fluorescence spectra in a compact, portable, and low-cost device. Fluorescence spectroscopy devices rely heavily on the precise acquisition of fluorescence spectra from biological tissues in both in vitro and in vivo measurements for diagnostic applications. The calibration and laboratory validation results demonstrate the efficacy of the developed device in capturing spectral signals, highlighting its potential use in oral and cervical cancer diagnosis. Future work will focus on conducting pilot clinical studies to evaluate the performance of the device in clinical settings and on developing robust AI/ML models for real-time cancer diagnosis.

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